

investors receive for trades in stocks receiving the lower tick size is likely to improve overall relative to the baseline due to a narrower quoted spread, even though the portion of their price labeled price improvement may decline.

One commenter suggested generally that the Commission use simulations to determine the optimal tick size without providing details about how such a simulation could be structured.¹³⁰⁴ However, it is unclear how such simulations would be structured, and the Commission is unaware of existing simulations or of existing frameworks for simulations studying the effect of tick sizes on market quality. Challenges associated with simulations include model accuracy and complexity, data quality, computational limitations, uncertainty and sensitivity, validation and verification, scalability, and the challenges associated with applying simulations to human behavior which is inherently unpredictable, and the challenges associated with modeling dynamic and evolving systems like financial markets.

One commenter, while not opposing a half-cent tick for some stocks, stated that volume on inverted exchanges implied that there may not be a demand to trade with significantly narrower tick sizes.¹³⁰⁵ Inverted venues, which as discussed in section VII.C.2.b, offer a rebate to liquidity demanders and charge a fee to liquidity providers effectively allow market participants to price orders within the tick by the amount of the rebate to liquidity demanders. The commenter states that if there were significant demand to trade within the tick size, then we would expect to see inverted exchanges capture significant market share among truly tick-constrained stocks.¹³⁰⁶

¹³⁰⁴ See Mitre Corp. Letter at 5.

¹³⁰⁵ See IEX Letter I at 14.

¹³⁰⁶ Id.

This argument, however, does not take into account the fact that inverted exchanges are less likely to be at the NBBO than maker-taker exchanges and thus the risk of an order not being able to be fulfilled on an inverted exchange is higher.¹³⁰⁷ If the inverted exchange cannot fill an order it would generally re-route it or cancel it, depending on the terms of the order. Both options are costly as re-routing orders usually involves a re-routing fee charged by the routing exchange and an access fee charged by the receiving exchange, which would make the trade more expensive to transact relative to just sending it first to a high volume maker-taker exchange. The transfer would also take a small amount of time which could increase adverse selection risk for the re-routed order. Failing to execute the order is also costly as it could expose the trader to increased costs if the market moves against the trader in the time it takes to submit a new order. Consequently, market share on inverted exchanges may be low for reasons other than the stock being tick-constrained. For these reasons, low volume on inverted exchanges is not sufficient to identify demand to trade within the quoted spread.

iii. Alternative definitions of tick-constrained

Some commenters supported using TWAQS to determine which stocks are tick-constrained.¹³⁰⁸ Other commenters stated that other criteria, in addition to or in place of the quoted spread was needed to identify truly tick-constrained securities.¹³⁰⁹ One commenter stated that adding additional criteria would decrease complexity as it would limit the number of stocks receiving a smaller tick size, and would prevent stocks from bouncing back and forth between

¹³⁰⁷ Id.

¹³⁰⁸ See, e.g., Budish Letter at 4 and Harris Letter at 7.

¹³⁰⁹ See, e.g., T. Rowe Price Letter at 4, NYSE Letter I at 3, BlackRock Letter at 5, Citadel Letter I at 30. Optiver Letter at 1, 2. See also infra section VII.F.1.a for additional analysis of these alternatives.

tick regimes.¹³¹⁰ The Commission agrees with commenters that the quoted spread is sufficient to determine whether a stock is tick-constrained as a matter of principle but acknowledges that not all stocks receiving the lower tick size will react in the same manner to the tick size. The Commission has considered commenters' views regarding variables other than quoted spread, and has conducted additional, supplemental analysis, presented in this section.

Commenters' suggestions for additional variables fell into three broad categories. The most common type of suggestion involved a depth measure, with commenters questioning whether a stock with a narrow-quoted spread was actually tick-constrained if the sizes of the quotes were small.¹³¹¹ Others suggested metrics based on trading volume.¹³¹² Another commenter suggested price as a measure of tick constraint.¹³¹³

Some commenters stated that even stocks with the narrow quoted spread may not benefit from a smaller tick unless other criteria were met.¹³¹⁴ For example, commenters stated that reducing the tick size for stocks that have narrow quoted spreads, but low depth, could lead to pennyning which could harm market quality.¹³¹⁵ Another commenter stated that if there is insufficient depth, a narrower tick size could harm market quality by fragmenting liquidity over

¹³¹⁰ See Citadel Letter II at 4. The Commission considers the effect of stocks moving between tick size regimes in section VII.D.1.d, *infra*.

¹³¹¹ See, e.g., Invesco Letter at 3 suggesting that we consider whether a stock had multiple bids and offers at the NBBO. BlackRock Letter at 5 suggesting we consider Average Quoted Size. Citadel Letter I at 30 suggesting we consider average size at the NBBO/Daily volume. Cboe, State Street, et al. Letter at 2 and BlackRock Letter at 5 suggesting we consider quote size at the NBBO to average trade size, NYSE Letter I at 3 suggesting a quote stability measure of how long it takes the NBBO to return to pre-trade levels after a trade takes liquidity at the NBBO. T. Rowe Price Letter at 4 suggesting that depth could be measured by how many exchanges are currently quoting the NBBO.

¹³¹² See, e.g., BlackRock Letter at 5, Citadel Letter I at 30 (both suggesting trading volume as a measure of tick-constrained) and T. Rowe Price Letter at 4 (suggesting a turnover based measure of tick-constrained).

¹³¹³ See BlackRock Letter at 5.

¹³¹⁴ See, e.g., Citadel Letter I at 5-6 and BlackRock Letter at 5, Invesco Letter at 3, Virtu Letter II at 17.

¹³¹⁵ See, e.g., Citadel Letter I at 5-6 and BlackRock Letter at 5.

multiple price levels – even if quoted spreads are tight.¹³¹⁶ Another commenter didn’t articulate specific mechanism by which harms that might arise due to failing to take into account other criteria, but encouraged the Commission to take a “pragmatic approach” starting with a narrow set of “truly tick-constrained stocks” as defined by multiple criteria.¹³¹⁷

As discussed above, quoted spread reflects supply and demand forces in the market for liquidity provision. Many of the concerns of commenters regarding a smaller tick would be expressed by wider quoted spreads; for example, pennyng could potentially reduce incentives for liquidity provision, raising quoted spreads. This argues in favor of quoted spread as the main criterion. Furthermore, it is unlikely that a stock which trades near \$0.01 nearly all the time does so at random and would continue to do so even if unconstrained. It is more likely that a stock trading with a quoted spread at the minimum pricing increment nearly all the time does so because the stock would trade with a narrower spread if that was possible, but the minimum quoting increment constrains the spread. Finally, quoted spread is correlated with depth, volume, and price.¹³¹⁸ For these reasons, it is likely that stocks with narrower quoted spreads will also have relatively high depth, and volume, and have lower prices. However, to fully address commenters’ concerns, the Commission has provided additional, supplemental analysis that examines whether there is meaningful variation of the effect of a reduction of tick size within those stocks that have lower quoted spreads.

¹³¹⁶ See Invesco Letter at 3.

¹³¹⁷ See Cboe Letter II at 2, 6.

¹³¹⁸ Using WRDS Intra Day Indicators Data for all stocks priced over \$5.00 without sym_suffix, the correlation between daily log quoted spread, log share volume, log price, and log average depth at the NBBO is -.39, .35, and -.49 respectively. See also Harris Letter at 7.

Specifically, the analysis presented in Table 9 explores the question of whether stocks with similarly narrow quoted spreads respond differently to the TSP tick size change when conditioning on other factors suggested by commenters, such as depth, volume, or price. In sum, our analysis does not find evidence supporting the hypothesis that a narrower tick size is likely to harm stocks when considering criteria such as depth, volume, or price in addition to the quoted spread. Instead, our analysis suggests that across most market quality metrics stocks with high and low price, volume, or depth respond in a similar magnitude and direction to the same tick size change. For some market quality metrics, one subset of stocks may not have a statistically significant response to the tick size change. However, in no cases do we observe that stocks with high or low price, volume, or depth respond in opposite directions with magnitudes that are statistically significant. Thus, we find no evidence of a benefit from adding criteria to the tick size determination that would compensate for the considerably greater complexity that such an addition would cause.

The analysis presented in Table 9 builds on the TSP analysis considered above, with a few modifications.¹³¹⁹ First, for brevity, we only report findings for bin 1 and bin 2 stocks – i.e., those stocks with narrower quoted spreads – because the final Rule will similarly only affect stocks with less than 2 ticks intra-spread.¹³²⁰ We also only present quantile regressions. We obtain daily regular hours trading volume, value weighted average price, and depth at the NBBO information from WRDS Intraday indicators for all trading days in May and June 2018. We

¹³¹⁹ See supra section VII.D.1.b.ii.

¹³²⁰ The Proposing Release analysis divided stocks into four bins based on their prevailing quoted spreads prior to the conclusion of the TSP. The first bin is for stocks with quoted spreads (\$0.00, \$0.06). The second bin is for stocks with quoted spreads in the range (\$0.06, \$0.09). The third bin is for stocks that had quoted spreads of (\$0.09, \$0.15) or approximately 2-3 ticks intra at a \$0.05 tick increment. The fourth bin is for stocks with quoted spreads greater than \$0.15.

divide bin 1 and bin 2 stocks into quartiles based on their average volume, price, or depth at the NBBO. The analysis uses quantile (median) regressions¹³²¹ to estimate the following difference-in-difference regression model separately for the high and low quartile samples:¹³²²

$$Y_{j,t} = \alpha_0 + \alpha_P Pilot_j + \alpha_E Event_t + \beta(Pilot_j \times Event_t) + \mu_{j,t}$$

By checking to see if market quality measures are affected differently for stocks with similarly narrow quoted spreads, but which have different depth, or price, or volume profiles, the analysis can show whether the effects of reducing the tick size depend on these additional characteristics in a manner that a quoted spread measure misses. The results are presented in Table 9. Panel A presents the analysis that considers the effect of trading volume on a stock's response to a reduction in the tick size. The analysis sorts and subdivides the treated and control stocks separately by volume into quartiles, and it then performs regressions on the top and bottom quartile of stocks separately. Columns one and two present the results for stocks with TWAQS less than or equal to \$0.011, whereas Columns three and four present the results for stocks with TWAQS greater than \$0.011 but less than or equal to \$0.020. Columns one and three present the results for stocks in the bottom quartile of trading volume while columns two and four present the results for stocks in the highest quartile of trading volume. Panel B presents the same analysis as Panel A but where stocks are sorted by price, and Panel C does the same for

¹³²¹ The primary advantage of quantile regressions is that they are less sensitive to outliers that can affect mean inference in OLS. Thus, median regressions provide additional robustness to the analysis and ensure that results are not driven by outliers.

¹³²² In this equation the variable Y denotes the response variable of interest such as quoted spread and depth. The subscripts j and t serve to index stocks and days respectively. α_0 , α_p , α_e , and β are coefficients (to be estimated), and $u_{j,t}$ is the error term. $Pilot_j$ is an indicator variable that equals 1 if stock j was in the treatment group, or 0 if stock j was in the control group. $Event_t$ is an indicator variable which is equal to 1 if the day t was post the treatment event and equals 0 otherwise. Table 8 reports the difference-in-difference estimator of β for a different response variable Y across the different quoted spread bins.

depth at the NBBO. We consider the differential effect of the TSP on depth, quoted spreads, effective spreads, cancel-to-trade ratio, share of odd-lot volume, and realized spread.

Table 9: TSP Analysis Conditional on Volume, Price, and Depth at the NBBO
Panel A: Trading Volume

Quoted Spread Bin # Volume	TWAQS $\leq$ \$0.06		\$0.06 < TWAQS $\leq$ \$0.09	
	Low	High	Low	High
Depth (100 shares)	-0.95*** (-11.81)	-1.24*** (-22.45)	-0.59*** (-8.58)	-0.87*** (-16.01)
Depth (\$1,000)	-9.66*** (-5.03)	-24.9*** (-8.89)	-4.54*** (-5.10)	-12.8*** (-7.43)
Quoted Spread (\$)	-0.021*** (-4.51)	-0.037*** (-23.01)	-0.0097 (-0.72)	-0.035*** (-6.57)
Relative quoted Spread	-0.0055*** (-3.71)	-0.0044*** (-7.58)	-0.00064 (-0.39)	-0.0012 (-1.41)
Effective spread (\$)	-0.028 (-1.24)	-0.039*** (-3.49)	0.015 (0.29)	0.027 (0.76)
Relative effective spread	-0.0037 (-0.92)	-0.0049*** (-3.43)	0.0079 (0.83)	0.00059 (0.91)
Cancel-to-trade	-1.31 (-0.59)	9.54*** (6.67)	6.64** (2.35)	7.58*** (6.45)
Share of odd-lot volume	0.00058** (2.07)	0.0018*** (8.82)	0.00068 (1.45)	0.0020*** (5.21)
Realized Spread	-0.013*** (-7.27)	-0.013*** (-23.69)	-0.0049 (-1.13)	-0.0086*** (-7.39)
Relative Realized Spread	-0.0034*** (-5.59)	-0.0016*** (-5.34)	-0.000061 (-0.08)	0.000100 (0.29)

Panel B: VWAP

Quoted Spread Bin # Price	TWAQS $\leq$ \$0.06		\$0.06 < TWAQS $\leq$ \$0.09	
	Low	High	Low	High
Depth (100 shares)	-1.13*** (-12.70)	-1.15*** (-30.31)	-0.56*** (-6.22)	-0.70*** (-24.81)
Depth (\$1,000)	-12.0*** (-5.81)	-19.3*** (-11.28)	-2.74* (-1.87)	-10.6*** (-10.04)
Quoted Spread (\$)	-0.026*** (-7.76)	-0.039*** (-10.35)	-0.012 (-1.21)	-0.031*** (-6.21)
Relative quoted Spread	-0.010***	-0.0014***	-0.0012	-0.00048***

	(-6.65)	(-4.50)	(-0.61)	(-4.74)
Effective spread (\$)	-0.0064***	-0.056***	-0.013	-0.0020
	(-5.70)	(-3.79)	(-0.24)	(-0.05)
Relative effective spread	-0.0030	-0.0026***	0.0053	-0.00083
	(-0.99)	(-4.09)	(0.57)	(-0.80)
Cancel-to-trade	-1.90	11.4***	4.87*	6.66***
	(-0.94)	(8.21)	(1.81)	(6.11)
Share of odd-lot volume	0.00058***	0.0029***	0.00076	0.0013***
	(3.05)	(9.98)	(1.55)	(3.42)
Realized Spread	-0.016***	-0.014***	-0.0062	-0.011***
	(-14.61)	(-22.63)	(-1.59)	(-11.31)
Relative Realized Spread	-0.0057***	-0.00066***	-0.00035	-0.00022***
	(-9.15)	(-16.55)	(-0.38)	(-12.28)

Quoted Spread Bin # Depth at the NBBO	Panel C: Depth at NBBO			
	TWAQS ≤ \$0.06		\$0.06 < TWAQS ≤ \$0.09	
	Low	High	Low	High
Depth (100 shares)	-1.02***	-1.38***	-0.62***	-0.76***
	(-20.53)	(-18.62)	(-21.27)	(-7.94)
Depth (\$1,000)	-10.4***	-26.4***	-7.67***	-9.26***
	(-9.46)	(-7.33)	(-16.76)	(-3.39)
Quoted Spread (\$)	-0.031***	-0.032***	-0.014**	-0.021**
	(-6.24)	(-15.74)	(-2.44)	(-2.11)
Relative quoted Spread	-0.00072	-0.0098***	-0.00015	-0.0026
	(-1.15)	(-8.28)	(-0.55)	(-1.27)
Effective spread (\$)	-0.038***	-0.031**	0.0049	-0.021
	(-3.37)	(-2.11)	(0.41)	(-0.39)
Relative effective spread	-0.0022***	-0.0078**	-0.00020	0.0055
	(-5.18)	(-2.05)	(-1.47)	(0.60)
Cancel-to-trade	5.83***	1.81	4.79***	6.10**
	(4.35)	(1.25)	(3.03)	(2.50)
Share of odd-lot volume	0.0028***	0.00097***	0.0014***	0.0015***
	(9.02)	(4.66)	(3.42)	(3.42)
Realized Spread	-0.012***	-0.016***	-0.0072***	-0.0082**
	(-8.60)	(-21.26)	(-5.60)	(-2.33)
Relative Realized Spread	-0.00074***	-0.0046***	-0.00013*	-0.00046
	(-5.69)	(-7.87)	(-1.76)	(-0.50)

* Panel A presents results for the sample sorted on trading volume, Panel B presents the results for price, and Panel C presents the results for depth at the NBBO. Within each bin, stocks are sorted based on their average trading volume, price, and depth at the NBBO during May and June 2018. Each regression is estimated using quantile (median) regressions. The 3 tables – one for each panel – present the effects of a reduction in minimum tick size from \$0.05 to \$0.01 cent on various quoting and trading outcome variables. A difference-in-difference regression with no control variables is estimated using data covering Control, Test Group 2, and Test Group 3 TSP stocks from 08/01/2018 – 11/30/2018. All observations are at the stock day level. The same model is used for all outcome variables. For each outcome variable Y_{jt} , the table presents only the difference in difference coefficient estimates

that indicate the effect of the TSP on the dependent variable. The outcome variables are: Depth (\$1,000) is dollar value of depth at the NBBO measured in \$1,000s. The quoted spread refers to the distance between the NBBO midpoint and the NBBO quote. The effective spread is the distance between the NBBO midpoint and the realized trade price; the realized spread is the distance between a future NBBO midpoint (5-minutes ahead) and the trade price. Relative spread measures are calculated as the spread scaled by the NBBO midpoint. The cancel-to-trade ratio is the daily number of order cancellations divided by the number of trades, for displayed orders. The odd-lot rate is the percentage of trades in a day which executed against an odd-lot quote. Share of odd-lot volume is the odd-lot rate reported in WRDS Intra Day Indicators and equals $100 * (\text{Odd lot Messages}) / (\text{Trades for Odd lots})$. Similar to the analysis in Table 8, stocks are divided into quoted spread bins based on average quoted spreads during May-June of 2018. Results are presented for only bin 1 and bin 2 stocks; see *supra* note 1268 for details on bin assignment. Regressions on the highest and lowest quartiles within each bin are performed; cells are darkly shaded if the regression estimates differ in sign between the highest and lowest quartiles within a bin, and cells are lightly shaded if the regression estimates agree in sign and in statistical significance between the highest and lowest quartiles within a bin. Depth (100 shares) is number of shares available at the NBBO in 100s. All data are Winsorized at the 1% and 99% level. The numbers in the () parentheses reflect t-statistics that are based on two-way stock-and-date clustered standard errors. Symbols *, **, and *** reflect statistical significance at 10%, 5%, and 1% type-1 error levels.

The results show that in most cases that there is relatively little disagreement in terms of how low and high characteristic stocks respond to the TSP. In Panel A, which presents results sorted based on trading volume, 10 out of the 20 pairs of regressions agree on both sign and statistical significance. A pair refers to the high and low quartile group for the same market quality outcome and the same quoted spread group – *e.g.*, columns one and two are a pair, and three and four are a pair. This means that in 10 out of the 20 regression pairs run, high and low trading volume stocks had market quality responses to the change in the tick size that were in the same direction and were statistically different from zero. A further 8 out of the 20 pairs agree on sign, but not statistical magnitude. This means that while the point estimators from the regressions indicate that both high and low volume stocks had market quality responses in the same direction, either one or both groups experienced an effect that could not be measured to be statistically different from zero.¹³²³ Lastly, only 2 out of 20 pairs disagree on sign, but there are

¹³²³ A result that is not statistically significant does not directly imply that there is no effect. Rather it implies that the test did not find enough evidence to overturn the hypothesis of no effect. This could be because 1) there is actually no effect, or 2) because the test did not have sufficient power to identify an existing effect.

no cases where both effects reject the hypothesis of no effect. This means that while the coefficients for the effect of the reduction in the tick size on high and low volume stocks disagree in the direction of the effect, in at least one case the effect is not statistically distinguishable from zero.

The story is similar for Panel B which presents results for price. 11 out of 20 regression pairs agree on sign and statistical significance. A further 7 out of 20 agree on the sign of the effect though not necessarily statistical significance while only 2 out of 20 disagree on sign, but again in these cases neither result is statistically different from zero. Lastly in Panel C, which presents results for the sort based on depth at the NBBO, 14 out of the 20 regression pairs agree on sign and statistical magnitude, 4 out of 20 agree on sign, but not statistical magnitude, and only 2 pairs disagree on sign, but in these cases neither regression provides a statistically significant result.

This analysis does not support the idea that failing to include price, volume, or depth-based criteria when determining which stocks receive a smaller tick size is likely to produce significant harm. In most cases, the analysis suggests that market quality is likely to respond similarly to a reduction in the tick size for stocks that are both high or low volume, price, or depth. It is possible that in some cases, some of the positive anticipated effects discussed above may be mitigated for certain subgroups of stocks. We fail to find evidence that certain subgroups of stocks will be significantly harmed by the amendments when only considering the time weighted quoted spread when assigning tick sizes.

c. Effect on Message Traffic and Market Data

One commenter stated that, since “reducing the minimum pricing increments for quotations” will result in “significantly more quotation increments within a penny-wide spread